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Erinacine S, a Rare Sesterterpene from the Mycelia of *Hericium erinaceus*

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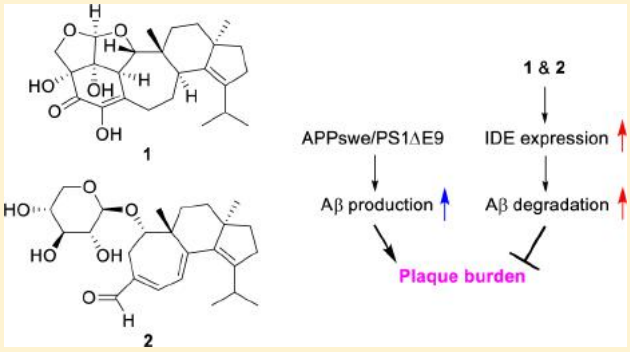
Supporting Information

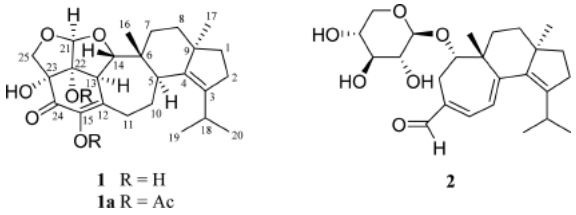
ABSTRACT: A new sesterterpene, erinacine S, and one cyathane diterpene xyloside, erinacine A, were isolated from the ethanol extract of the mycelia of *Hericium erinaceus*. Their structures were elucidated by spectroscopic and X-ray analysis. A 30-day oral course of erinacines A and S attenuated Aβ plaque burden in the brains of 5-month-old female APP/PS1 transgenic mice. Moreover, erinacines A and S significantly increased the level of insulin-degrading enzyme in cerebral cortex.



Alzheimer's disease (AD), an age-related and progressive neurodegenerative disorder, is characterized by the formation of neurofibrillary tangles, deposition of amyloid-β (Aβ) peptide in senile plaques, and neural and synaptic loss.[1] Aβ is a product from the proteolytic processing of amyloid precursor protein (APP) by β- and γ-secretases. The accumulation of Aβ results in neuroinflammation, oxidative stress, and Aβ deposition. Aβ can be degraded by several enzymes, such as insulin-degrading enzyme (IDE) and neprilysin (NEP). Although the pathogenesis of AD remains unknown, many studies support AD being caused by the imbalance or insufficiency of neurotrophic factors, which have been indicated to possess the ability to prevent neural and synaptic loss and modulate APP procession in AD.[2]

Hericium erinaceus, known as the lion's mane mushroom, is a basidiomycete mushroom, which was used in Chinese folk medicine to treat the tumors of the digestive system, such as esophageal, stomach, and duodenum cancers.[3,4] A group of cyathin diterpenoids (erinacines A−K, P, Q, and R) were isolated from the cultured mycelia of *H. erinaceus*.[5] In previous studies, it was shown that erinacines A−H exhibited a potent stimulating effect on nerve growth factor (NGF) synthesis *in vitro*.[6−12] Erinacine A was also reported to show a strong enhancing effect on increasing the content of catecholamine and NGF in the central nervous system of rats.[13] The NGF synthesis stimulators may be potential drug candidates for central nervous system disorders that are due to degeneration of neurons, such as Alzheimer's disease.[14,15] In our continuing search for bioactive components from the mycelia of *H. erinaceus*, we found a novel sesterterpene, erinacine S (**1**), and the known cyathin diterpene xyloside erinacine A (**2**). We also investigated the effect of erinacines A and S on amyloid pathology in 5-month-old APP/PS1 mice. In this report, the isolation and structure elucidation of **1** and the effect of **1** and **2** on amyloid are described.

The mycelia of *H. erinaceus* were refluxed with 95% ethanol, and the ethanol extract was partitioned with H_2O−EtOAc (1:1) to afford an H_2O layer and an EtOAc layer. The EtOAc layer was repeatedly chromatographed on silica gel and Sephadex LH-20 to give a new sesterterpene, erinacine S (**1**), and a known diterpenoid, erinacine A (**2**). The structure of **2** was confirmed by comparison of its ^{1}H and ^{13}C NMR data with those reported in the literature.[6,12]



Compound **1** was obtained as colorless crystals. Its molecular formula was determined to be $C_{25}H_{34}O_6$ by HRESIMS. The ^{13}C and DEPT NMR spectra (Table 1) displayed four methyl, seven methylene, five methine, and nine nonprotonated carbon signals.

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Table 1. NMR Spectroscopic Data (600 MHz, pyridine-d_5) for Erinacine S (1)[a]

position	δ_C, type	δ_H (J in Hz)	HMBC[a]
1	38.2, CH_2	1.54, dt (13.2, 7.8)	2, 3, 4, 8, 9, 17
		1.61, dt (13.2, 7.8)	
2	28.4, CH_2	2.28, t (7.8)	1, 3, 4, 9
3	138.6, C		
4	139.3, C		
5	46.5, CH	2.64, d (9.6)	3, 4, 6, 9, 10, 11, 16
6	42.5, C		
7	28.9, CH_2	1.80, m	5, 6, 8, 9, 16
8	36.8, CH_2	1.40, m	1, 7, 9, 17
		1.51, dd (12.6, 6.0)	
9	49.6, C		
10	25.1, CH_2	1.69, m	4, 5, 6, 11, 12
		2.02, m	
11	32.3, CH_2	1.99, m	12, 13, 15
		3.76, m	
12	129.4, C		
13	49.1, CH	3.80, dd (10.8, 1.8)	6, 12, 14, 15, 22, 23
14	91.3, CH	4.40, d (10.8)	6, 7, 12, 16
15	142.7, C		
16	19.4, CH_3	1.11, s	5, 6, 7, 14
17	25.0, CH_3	1.07, s	1, 4, 8, 9
18	27.3, CH	2.77, septet (6.6)	2, 3, 4, 19, 20
19	21.4, CH_3	0.97, d (6.6)	3, 18, 20
20	21.9, CH_3	0.98, d (6.6)	3, 18, 19
21	113.3, CH	6.13, s	13, 14, 25
22	93.0, C		
23	83.2, C		
24	193.6, C		
25	80.8, CH_2	4.48, d (10.2)	21, 22, 23, 24
		4.60, d (10.2)	
15-OH		10.76, s	12, 15, 24

[a]HMBC corrections, optimized for 8 Hz, are from proton(s) stated to the indicated carbon.

The signal at δ_C 193.6 was derived from a carbonyl group. Four nonprotonated carbon signals (δ_C 129.4, 138.6, 139.3, and 142.7) were in the olefinic region, suggesting two tetrasubstituted double bonds in compound **1**. Besides, two nonprotonated carbons (δ_C 83.2 and 93.0) in the aliphatic region were oxygenated. The one-bond $^1H-^{13}C$ connectivities were analyzed by using HSQC data. Four methyl carbons resonating at δ_C 19.4, 21.4, 21.9, and 25.0 correlated to protons at δ_H 1.11 (s), 0.97 (d, J = 6.6 Hz), 0.98 (d, J = 6.6 Hz), and 1.07 (s), respectively. Moreover, one methine carbon at δ_C 91.3 and one methylene carbon at δ_C 80.8 correlated to protons at δ_H 4.40 (d, J = 10.8 Hz) and 4.48 (d, J = 10.2 Hz), 4.60 (d, J = 10.2 Hz), respectively, which revealed that one methine and one methylene carbon were oxygenated. The COSY spectrum indicated five isolated spin systems as follows: H_2-1/H_2-2; H-5/H_2-10/H_2-11; H_2-7/H_2-8; H-13/H-14; H_3-19/H-18/H_3-20. Further connectivities were established by a long-range HMBC experiment, and the correlations are shown in Table 1. The HMBC cross-peaks of H-1/C-2, C-3, C-4, C-8, C-9, C-17, H-5/C-3, C-4, C-6, C-9, C-10, C-11, C-16, H_2-7/C-5, C-6, C-8, C-9, C-16, H_2-10/C-4, C-5, C-6, C-11, C-12, H-13/C-6, C-12, C-14, C-15, H-14/C-6, C-7, C-12, C-16, and H-18/C-2, C-3, C-4, C-19, C-20 suggested the presence of a fused 5−6−7 tricarbocyclic cyathane moiety in compound **1**. The HMBC data also showed cross-peaks of H-13/C-22, C-23, H-21/C-13, and H_2-25/C-22, C-23, C-24, indicating that one more isoprene unit was attached to the diterpene cyathane moiety and the rare sesterterpene carbon skeleton for **1** was established. The carbon−carbon connectivities were further confirmed by an INADEQUATE experiment, which showed all carbon−carbon connectivities except a double bond between C-3 and C-4. Moreover, proton H-21 at δ_H 6.13 (s) correlating to the carbon resonating at δ_C 113.3 showed HMBC correlations to C-14 and C-25, which indicated that C-21 was dioxygenated and linked to C-14 and C-25 via oxygen atoms as an acetal. The relative configuration of **1** was deduced from a NOESY experiment, which exhibited cross-peaks of H-5/H-13, H_3-17 and H-14/H_3-16. Thus, it was suggested that H-5, H-13, and H_3-17 were on the same face of **1**, in the α position, and H-14 and H_3-16 were in the β position. In order to determine the stereochemistry of C-22 and C-23, compound **1** was acetylated to give diacetate **1a** suitable for single-crystal X-ray crystallographic analysis; the perspective ORTEP plot is shown in Figure 1. The X-ray analysis confirmed the proposed structure for **1** and

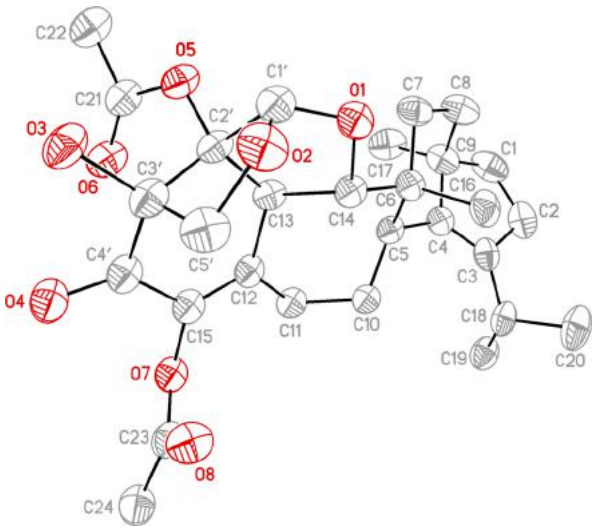

Figure 1. Stereo ORTEP plot of erinacine S 15,22-diacetate (**1a**).

indicated that the two oxygen atoms at C-22 and C-23 were in the α position. Accordingly, the structure of sesterterpene **1** was established, and it was given the trivial name erinacine S.

In order to investigate the potentials of **1** and **2** on amyloid pathology, 5-month-old APPswe/PS1ΔE9 (APP/PS1) double transgenic mice, a mouse model of Alzheimer's disease, were used. APP/PS1 mice have been found to exhibit impaired exploratory behavior, spatial memory, and synaptic function and increased Aβ production at the age of 5 months.[16] The administration of **1** and **2** reduced the AB10-stained plaque burden by 38.1 ± 19.7% and 40.2 ± 15.2%, respectively (Figure 2). The burden reducing effect was statistically significant after the treatment with **1** and **2**. The levels of nerve growth factor, insulin degrading enzyme, and neprilysine were also measured. The level of NGF was not significantly elevated after the treatment of **1** and **2** (data not shown). The treatment of **1** and **2** increased the level of IDE in the cortex by 141.1 ± 63.7% and 130.5 ± 68.9%, respectively. However, the level of NEP was not altered (Figure 3). Thus, it is suggested that **1** and **2** may reduce the AB10-stained plaque burden through increasing Aβ degradation by elevating the level of IDE, but not NEP or NGF.

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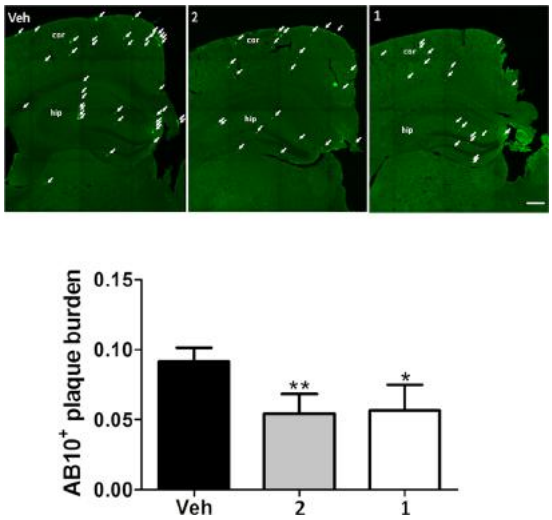

Figure 2. Effects of erinacines S (**2**) and A (**1**) on reducing amyloid plaque burden in the cerebral cortex of APP/PS1 mice. Five-month-old APP/PS1 mice were treated orally with vehicle (Veh), **2**, or **1** for 30 days. Amyloid plaques were detected by immunohistochemical staining with AB10 antibody. Panel A shows the representative fluorescent images of AB10 immunostaining plaque (arrow) in the area including the cerebral cortex (cor) and hippocampus (hip). Scale bar: 250 μm. Panel B shows the plaque burden in an AB10-stained semicerebral sphere calculated by image analysis software. Plaque burden is displayed as a percentage of the area occupied by AB-10-stained signal in the area of interest. The results are the mean ± SD. Significant differences between treated and control (Veh) groups are indicated by *, $p < 0.05$; **, $p < 0.01$.

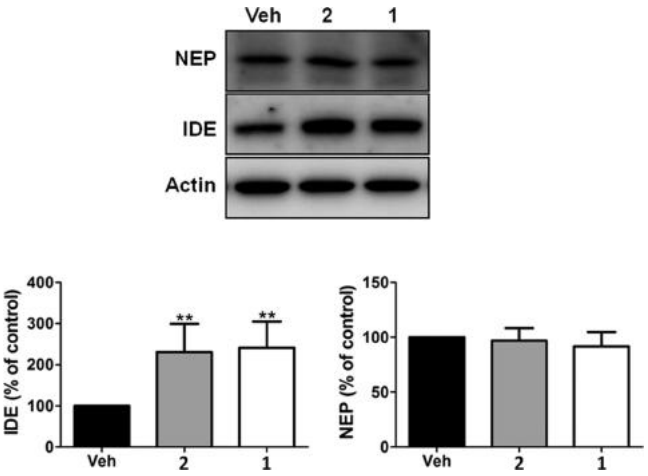

Figure 3. Effects of erinacines S (**2**) and A (**1**) on the levels of Aβ-degrading enzymes (IDE and NEP) in the cerebral cortex of APP/PS1 mice. Five-month-old APP/PS1 mice were treated orally with vehicle (Veh), **2**, or **1** for 30 days. Cortex was removed and homogenized, and the proteins Aβ-degrading enzymes in the homogenate were analyzed by immunoblotting. Representative immunoblots of IDE and neprilysin are shown (upper panel). The levels of IDE, neprilysin, and β-actin were examined by immunoblotting. The results are the mean ± SD. Significant differences between treated and control (Veh) groups are indicated by **, $p < 0.01$.

■ EXPERIMENTAL SECTION

General Experimental Procedures. Melting points were determined on a Yanaco MP-I3 micro melting point apparatus and are uncorrected. Optical rotation was obtained on a JASCO DIP-370 digital polarimeter. UV spectrum was measured on a Hitachi U-3310 spectrophotometer. ^{1}H, ^{13}C, and 2D NMR spectra were recorded on a Varian VNMRS 600 MHz spectrometer. ESIMS and HRESIMS were obtained on Finnigan MAT LCQ and JEOL JMS-700 spectrometers, respectively. X-ray crystal structure data were obtained on an Agilent Xcalibur Atlas Gemini single-crystal X-ray diffractometer. Silica gel 60 (Merck, Germany), Sephadex LH-20 (Amersham Biosciences, Sweden), and Cosmosil C18-OPN (Nacalai Tesque, Japan) were used for column chromatography.

Plant Material. *H. erinaceus* (BCRC 35669) was purchased from Bioresources Collection and Research and Development Institute (Hsinchu, Taiwan).

Cultivaton of *H. erinaceus*. An *H. erinaceus* agar slant was transferred onto a potato dextrose agar plate and incubated at 26 °C for 15 days. A fresh mycelium block (1 cm^2) was inoculated into a 2 L Erlenmeyer flask containing 1.3 L of modified medium (0.25% yeast extract, 4.5% glucose, 0.5% soybean powder, 0.25% peptone, and 0.05% $MgSO_4$; pH was adjusted to 4.5). The whole medium was incubated at 26 °C on a 120 rpm shaker for 5 days. The fermentation process was then scaled up from a 2 L shake flask to 500 L for 5 days.

Extraction and Isolation. The freeze-dried mycelia of *H. erinaceus* (2 kg) were refluxed with 95% ethanol. The ethanol solution was concentrated under vacuum to give a brown extract, which was partitioned with H_2O−EtOAc (1:1) to afford a H_2O layer and EtOAc layer. The EtOAc layer was chromatographed on a silica gel column (70−230 mesh, 70 × 10 cm) and eluted with a gradient system of *n*-hexane−EtOAc (10:1; 3:1; 3:2; 1:1; 1:2; 0:1) to give seven fractions (Fr. I−VII). Fraction III, the eluate of *n*-hexane−EtOAc (3:2), was rechromatographed on Sephadex LH-20 and silica gel columns to afford **1** (525 mg). Fraction VI, an eluate of *n*-hexane−EtOAc (1:2), was separated on a Sephadex LH-20 column eluting with MeOH to yield two subfractions (Fr. VI-1 and VI-2). Subfraction VI-2 was rechromatographed on Sephadex LH-20 (70% MeOH) and RP-18 (60% MeOH) columns to afford **2** (8.4 g).

Erinacine S (1): colorless prisms; mp 269−271 °C; $[\alpha]^{25}_{D}$ +47.7 (*c* 0.9, MeOH); UV (MeOH) λ_{max} (log ε) 201 (4.18), 284 (4.04) nm; ^{1}H and ^{13}C NMR data are shown in Table 1; ESIMS *m/z* 429 [M − H]$^-$; HRESIMS *m/z* 429.22719 (cacld for $C_{25}H_{33}O_6$, 429.22717).

Acetylation of 1. Compound **1** (20 mg) was treated with acetic anhydride (1 mL) in pyridine (1 mL) at room temperature. After stirring overnight, the reaction mixture was poured into ice water to give a precipitate. The precipitate was collected by filtration, washed with water, and recrystallized from CH_3OH−H_2O to give a diacetate product **1a** (17 mg): mp 154−156 °C; ^{1}H NMR ($CDCl_3$) δ 5.85 (1H, s, H-21), 4.50 (1H, d, *J* = 10.8 Hz, H-25), 4.16 (1H, d, *J* = 10.2 Hz, H-14), 4.02 (1H, d, *J* = 10.2 Hz, H-25), 3.13 (1H, d, *J* = 10.2 Hz, H-13), 3.06 (1H, m, H-11), 2.68 (1H, septet, *J* = 6.6 Hz, H_3-18), 2.28 (3H, m, H_2-2, H-5), 2.25 (3H, s, 15-OAc), 2.12 (3H, s, 22-OAc), 2.02 (1H, m, H-11), 1.99 (1H, m, H-10), 1.72 (1H, m, H-10), 1.62 (1H, m, H-1), 1.58 (1H, m, H-7), 1.56 (1H, m, H-1), 1.52 (1H, m, H-8), 1.38 (1H, m, H-7), 1.04 (3H, s, H_3-17), 0.99 (3H, s, H_3-16), 0.95 (3H, d, *J* = 7.2 Hz, H_3-19), 0.94 (3H, d, *J* = 7.2 Hz, H_3-20); ^{13}C NMR ($CDCl_3$) δ 188.6 (C-24), 171.0 (22-OCOCH_3), 168.7 (15-OCOCH_3), 143.2 (C-12), 139.3 (C-3), 138.0 (C-15), 137.4 (C-4), 112.0 (C-21), 94.8 (C-22), 89.8 (C-14), 81.0 (C-23), 80.6 (C-25), 49.2 (C-9), 47.6 (C-13), 45.9 (C-5), 41.3 (C-6), 38.1 (C-1), 36.4 (C-8), 32.1 (C-11), 28.2 (C-2), 27.9 (C-7), 27.1 (C-18), 24.9 (C-17), 22.0 (C-10), 21.8 (C-19 or C-20), 21.3 (C-19 or C-20), 20.6 (22-OCOCH_3), 20.2 (15-OCOCH_3), 18.6 (C-16).

Animal Handling. The Institutional Animal Care and Use Committee at the National Research Institution of Chinese Medicine approved the animal protocol (IACUC No: 100-A-04 and 102-417-3). The APP/PS1 transgenic mouse model of AD was purchased from Jackson Laboratory (No. 005864) and expressed a chimeric mouse/human APP695 harboring the Swedish K670M/N671L mutations (APPswe) and human PS1 with the exon-9 deletion mutation (PS1ΔE9). Breeding gender ratio was a male with two females in one cage. Experiments were conducted using wild-type siblings and AD transgenic female C57BL/6J mice. The animals were housed under controlled room temperature (24 ± 1 °C) and humidity (55−65%) with a 12:12 h (07:00−19:00) light−dark cycle. All mice were provided with commercially available rodent normal chow diet and water *ad libitum*. In

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the APP/PS1 transgenic mice model, plaques are known to be well established and visible at 6 months old. To explore the effects of **1** and **2**, 5-month-old APP/PS1 mice were fed with **1** or **2** (30 mg/kg/day) for 30 days.

Immunohistochemistry. After administration by oral gavage to 5-month-old mice for 30 days, mice were sacrificed by transcardial perfusion with saline at pH 7.4. Cerebral cortex was processed for immunohistochemical detection. Dissected brains were immersed in 4% formaldehyde for 18 h at 4 °C, then cryoprotected in sucrose before being sectioned into 30 μm thick free-floating sections. Sections were blocked in blocking buffer (phosphate buffer saline (PBS) with 3% normal donkey serum, 1% bovine serum albumin (BSA), and 0.3% Triton X-100) for 60 min and incubated overnight at 4 °C in antibody dilution buffer (PBS with 1% normal donkey serum, 1% BSA, and 0.3% Triton X-100) with mouse anti-Aβ1-16 antibody (AB10, 1:300, Millipore) and fluorescein isothiocyanate-conjugated donkey anti-mouse IgG at room temperature for 2 h. Sections were then washed with PBS containing 0.01% Triton X-100 and mounted with Aqua Poly/Mount (Polyscience Inc., Warrington, PA, USA). Fluorescent images were taken using a Zeiss LSM 780 confocal microscope (Jena, Germany). Amyloid plaque deposition was quantified using ImageJ software. The amyloid plaque burden indicates the AB10-reactive area normalized to the total area.

Immunoblots. For Western blot analysis, samples (30 μg of protein) were separated by sodium dodecyl sulfate-polyacrylamide gel electrophoresis and were then transferred to PVDF membranes. The primary antibodies used were as follows: rabbit anti-NGF antibody (Millipore), rabbit anti-NEP antibody (Millipore), rabbit anti-IDE antibody (Millipore), and mouse anti-β-actin antibody (Millipore). The secondary antibodies were anti-rabbit IgG antibody conjugated with horseradish peroxidase (HRP; GE Healthcare) and anti-mouse IgG antibody conjugated with HRP (Jackson ImmunoResearch). Enhanced chemiluminescence detection reagents (GE Healthcare) were used for detection. Bands were quantified using a Fujifilm LAS-3000 luminescent image analyzer (Tokyo, Japan).

Statistical Analysis. The results are expressed as the mean ± standard deviation and were interpreted by analysis of variance (ANOVA) with *post hoc* Bonferroni multiple comparisons tests.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jnatprod.5b00474.

NMR spectra for **1** and **1a** (PDF)
X-ray crystallographic data for **1a** (CIF)

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Notes

The authors declare no competing financial interest.

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